

Instruction Tuning

Agenda

- Motivation
 - General RLHF Approach
 - Where to get SFT data from?
 - PPO, GRPO, DPO
-
- Reasoning
 - DeepSeek R1

GPT for dialog systems

A model that can only generate continuations of text isn't very useful.

We want it to be able to carry on a dialogue.

GPT

– What's the weather like in Moscow tomorrow?

What's the weather like in Paris tomorrow? What's the weather like in London tomorrow?

ChatGPT

– What's the weather like in Moscow tomorrow?

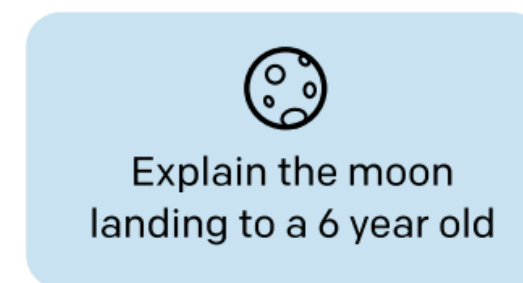
– Tomorrow in Moscow it will be +15°, partly cloudy.

Reinforcement Learning with Human Feedback

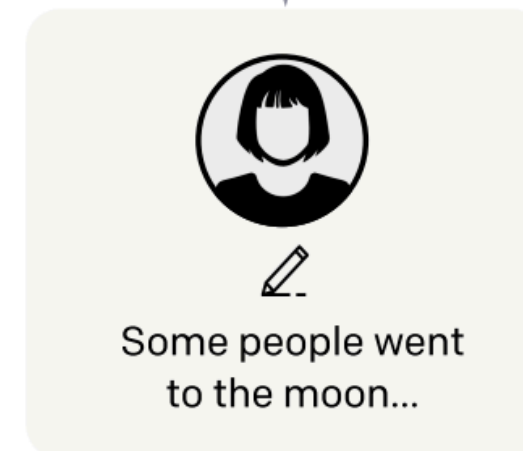
Step 1

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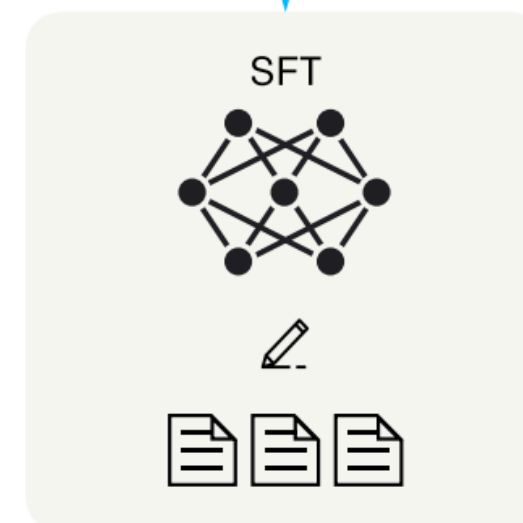
A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



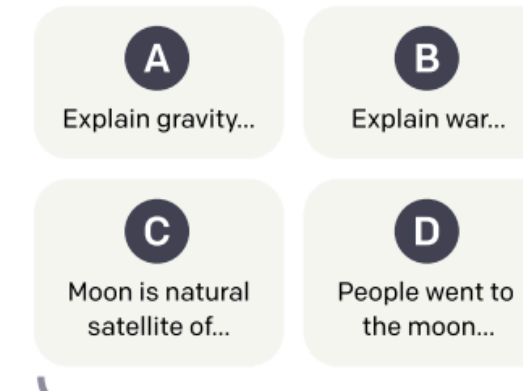
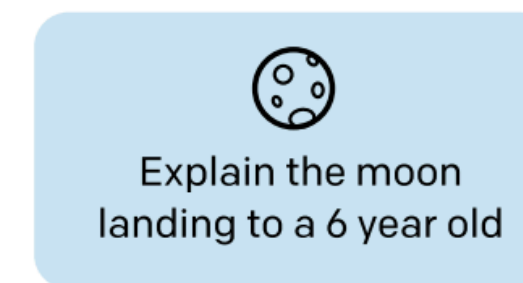
This data is used
to fine-tune GPT-3
with supervised
learning.



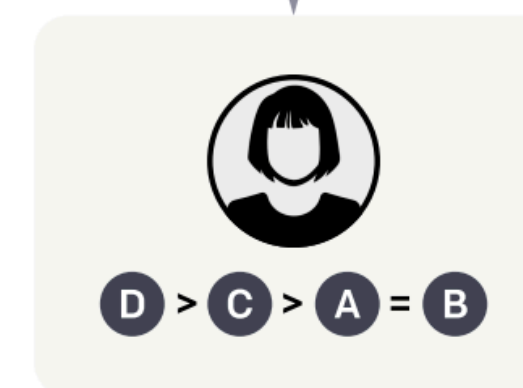
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**Collect comparison data,
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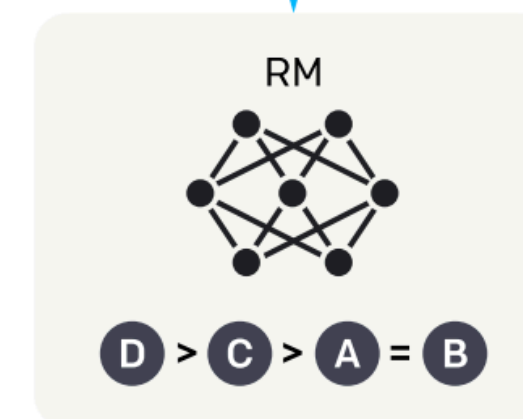
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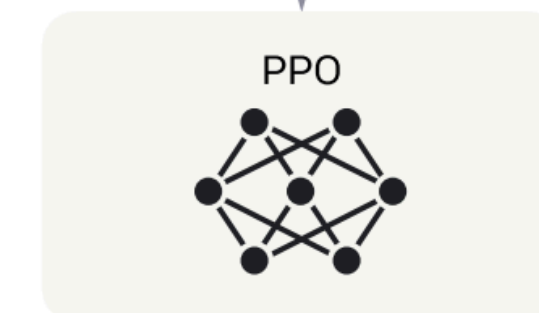
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the reward model using
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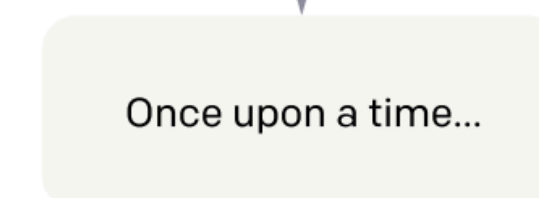
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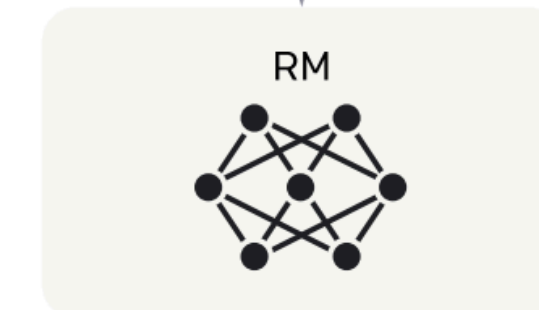
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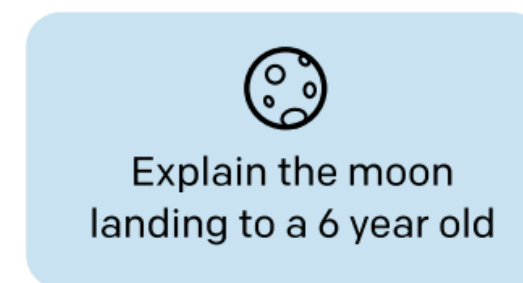


Reinforcement Learning with Human Feedback

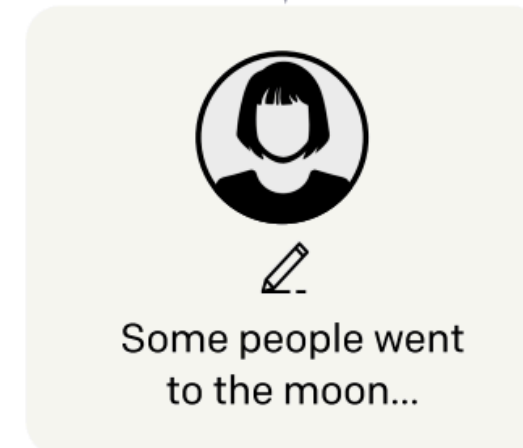
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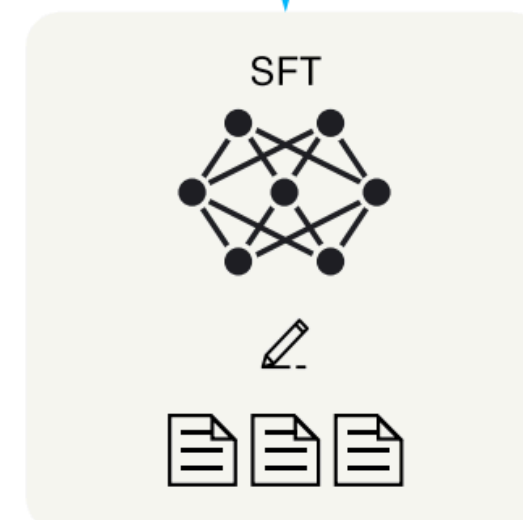
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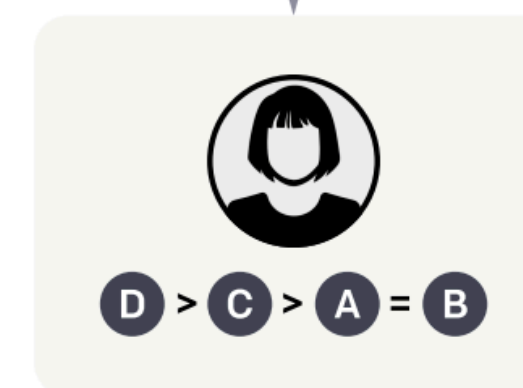


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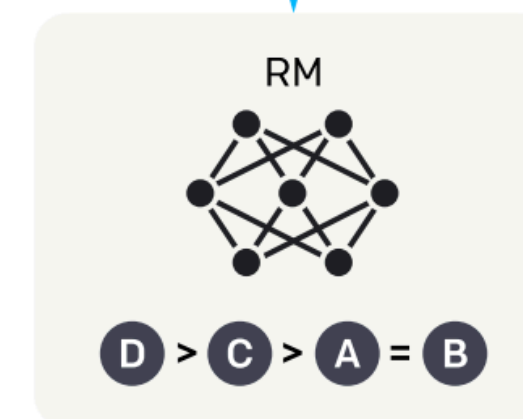
**Collect comparison data,
and train a reward model.**

- SFT requires manual labeling of large volumes of data.
- This is expensive and complex.
- Can it be done faster and cheaper?

A labeler ranks
the outputs from
best to worst.

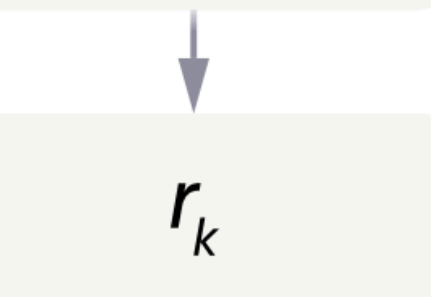
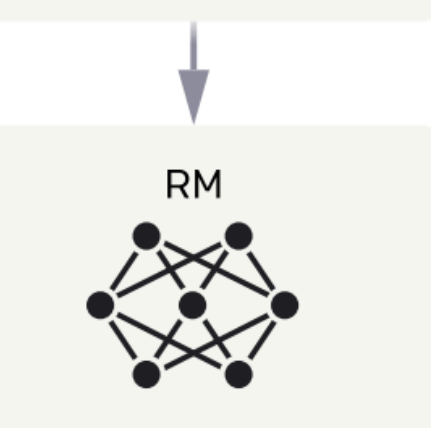
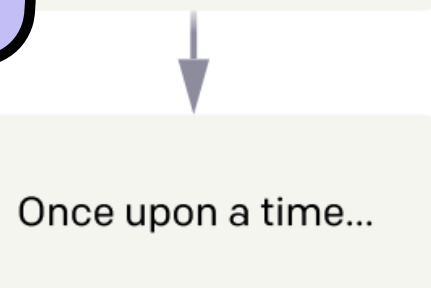
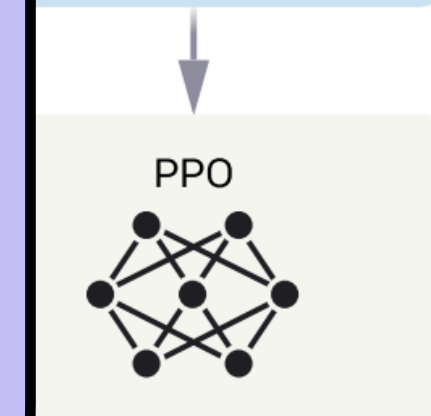


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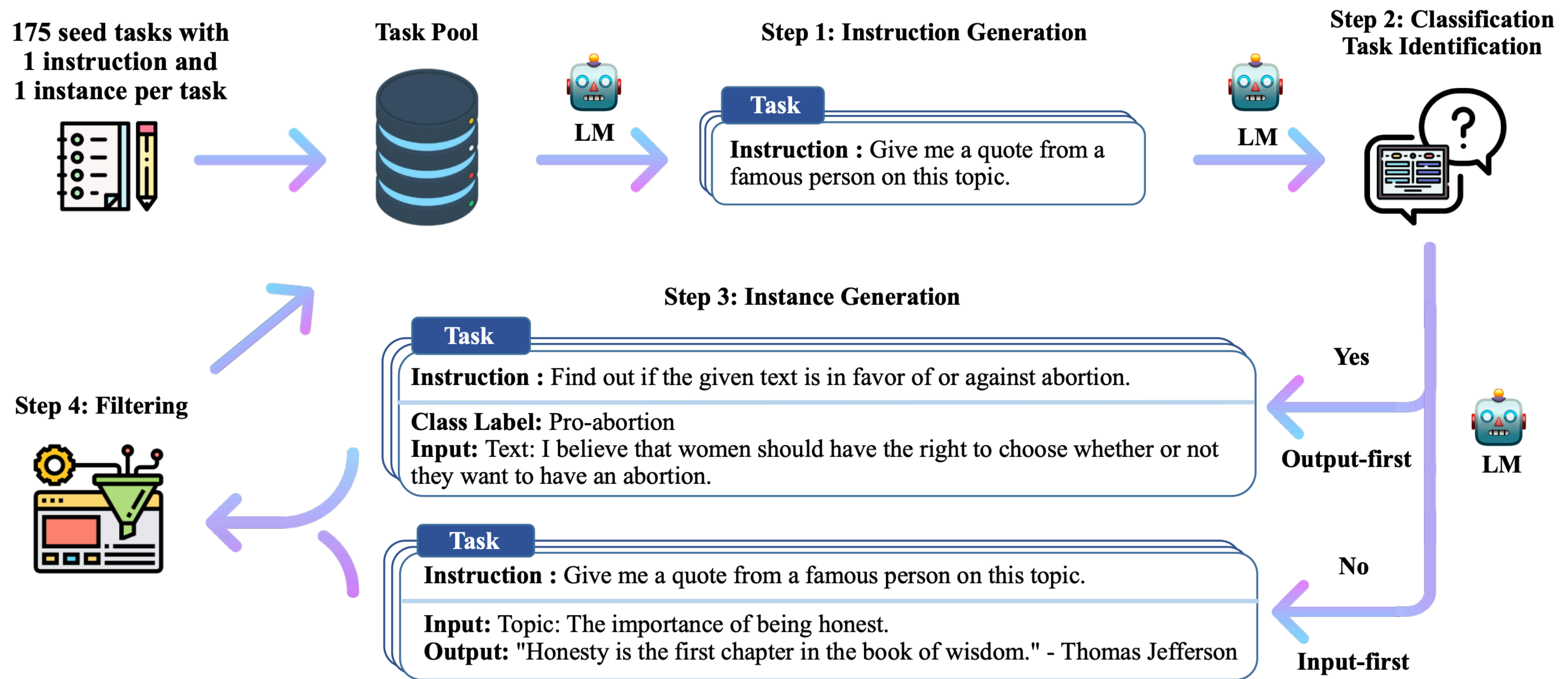
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Self-Instruct



Self-Instruct: instruction generation

Come up with a series of tasks:

1. I am planning a 7-day trip to Seattle. Can you make a **detailed** plan for me?
2. Is there anything I can eat for a breakfast that doesn't **include** eggs, yet **includes** protein, and has roughly 700-1000 calories?
3. Translate this sentence into any Arabic dialect and say **what dialect** it is: "The beautiful ones are not yet born
4. Given a set of numbers, find all possible subsets that **sum** to a **given** number.
5. Give me a phrase that I can use to express I am very **happy**.
6. Create a birthday planning checklist.
7. What is the relation between the **given** pairs?
8. Tell if the sentence contains toxic language.
9. Create a list of 10 words that have the letter "k" in them.
10. Create a list of all countries and their capitals from Africa.
11. I am looking for a job, but it's difficult to find one. Can you help me?
12. Find the longest word in this sentence: The quick brown fox jumps over the lazy dog.
13. Translate these numbers into English (word format): 3168987645561523
14. What is the next number in the series: 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096...?
15. Write a program that will tell me if a given string contains anagrams.

Submit

163

Mode

Model

davinci

Temperature0.7

Maximum length1024

Stop sequences

Enter sequence and press Tab

16. x

Top P0.5

Frequency penalty0

Presence penalty2

Best of1

Self-Instruct: instruction classification

We define the task: classification or generation

Can the following task be regarded as a classification task with finite output labels?

Task: Given my personality and the job, tell me if I would be suitable.
Is it classification? Yes

Task: Give me an example of a time when you had to use your sense of humor.
Is it classification? No

Task: Replace the placeholders in the given text with appropriate named entities.
Is it classification? No

...

Task: To make the pairs have the same analogy, write the fourth word.
Is it classification? No

Task: Given a set of numbers, find all possible subsets that sum to a given number.
Is it classification? No

Task: {instruction for the target task}

Self-Instruct: answer generation

Come up with examples for the following tasks. Try to generate multiple examples when possible. If the task doesn't require additional input, you can generate the output directly.

Task: Which exercises are best for reducing belly fat at home?

Output:

- Lying Leg Raises
- Leg In And Out
- Plank
- Side Plank
- Sit-ups

...

Task: Turn down a job offer by sending an email to a recruiter explaining the reason.

Output: Hi [Recruiter],

Thank you so much for the generous offer to join your team. As we discussed, I've admired the company for a number of years, and am a proud endorser of its products. However, after further consideration of where I currently am in my career, I've decided to accept an offer at another company.

I would love to stay in touch with you and have already started following you on [Social Media Platform]. Again, thank you so much for your time and consideration.

Thanks again,
[Your Name]

Task: {Instruction for the target task}

Self-Instruct: filtering

A new instruction is added to the database only if:

- ROUGE-L < 0.7 for any query in the database
- The query does not contain special words (image, graph)

Answers are filtered:

- By uniqueness
- By statistics (length, repetitions)

Self-Instruct: results

Human evaluation of data quality

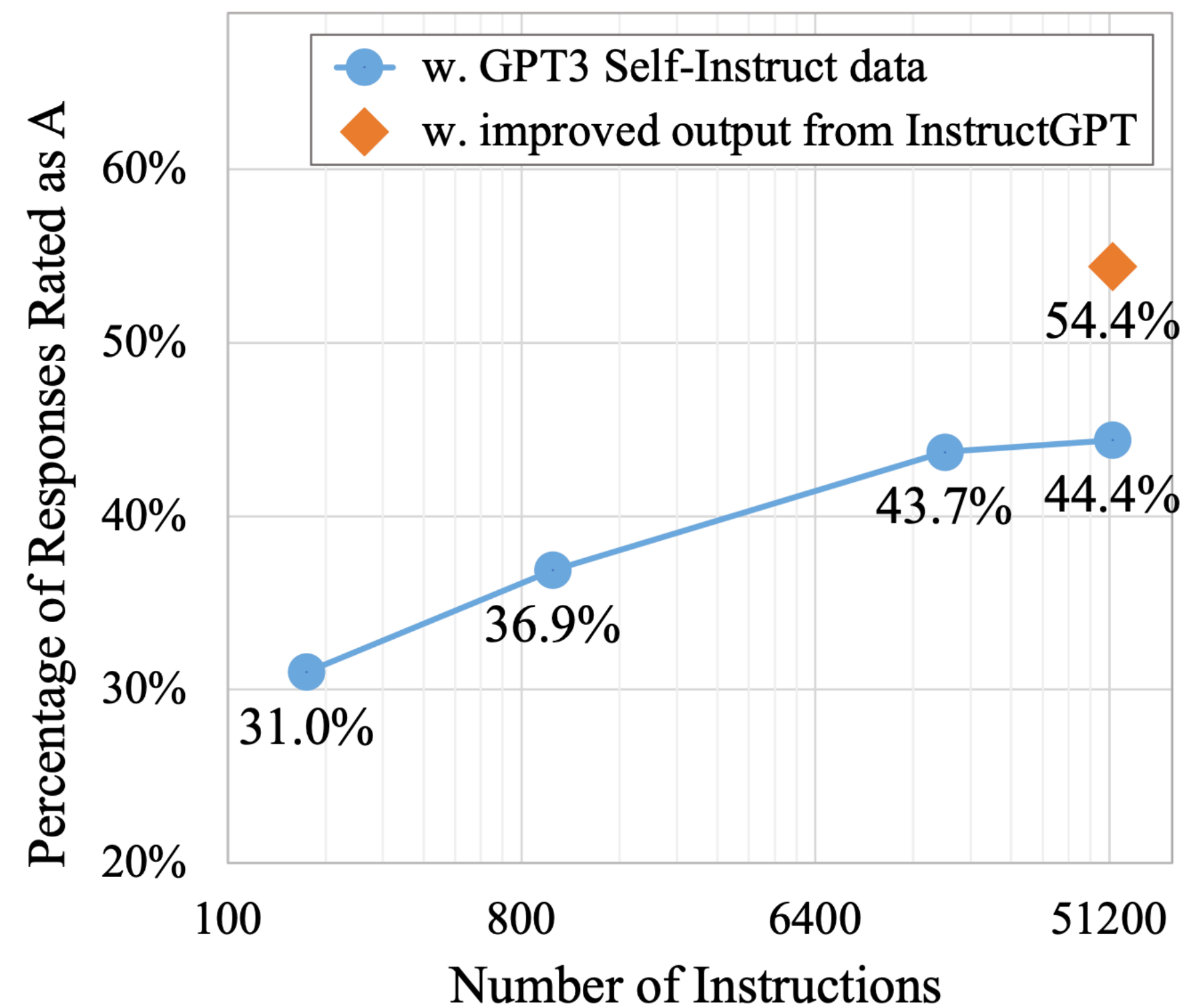
Quality Review Question	Yes %
Does the instruction describe a valid task?	92%
Is the input appropriate for the instruction?	79%
Is the output a correct and acceptable response to the instruction and input?	58%
All fields are valid	54%

Quality on unseen SuperNI examples

Model	# Params	ROUGE-L
Vanilla LMs		
T5-LM	11B	25.7
GPT3	175B	6.8
Instruction-tuned w/o SUPERNI		
T0	11B	33.1
GPT3 + T0 Training	175B	37.9
GPT3 _{SELF-INST} (Ours)	175B	39.9
InstructGPT ₀₀₁	175B	40.8
Instruction-tuned w/ SUPERNI		
Tk-INSTRUCT	11B	46.0
GPT3 + SUPERNI Training	175B	49.5
GPT3 _{SELF-INST} + SUPERNI Training (Ours)	175B	51.6

Self-Instruct: results

SFT doesn't require a lot of data

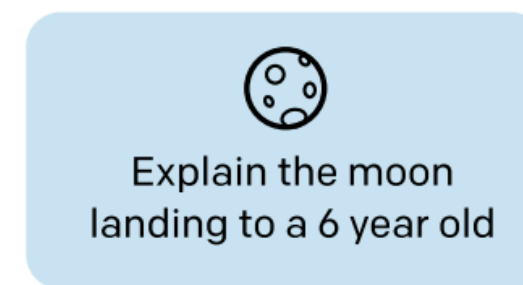


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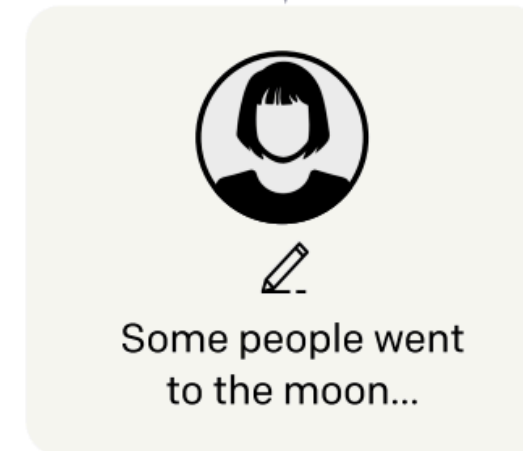
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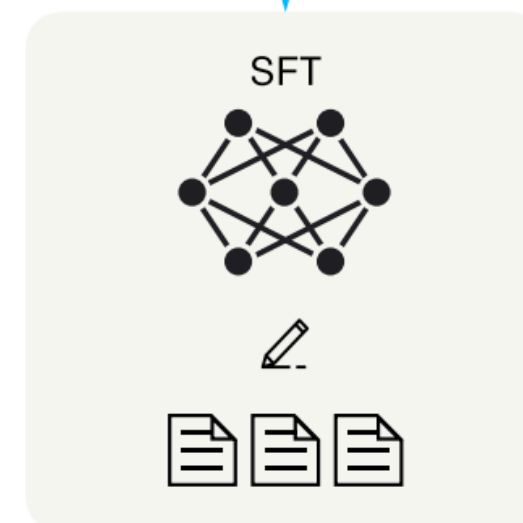
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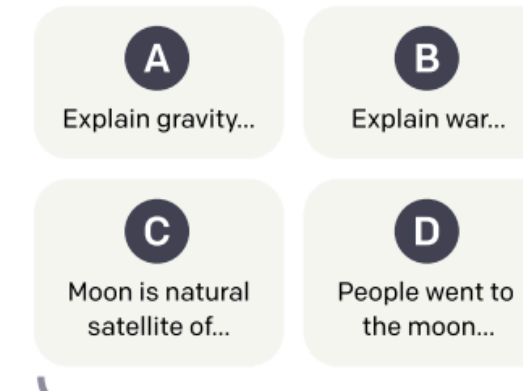
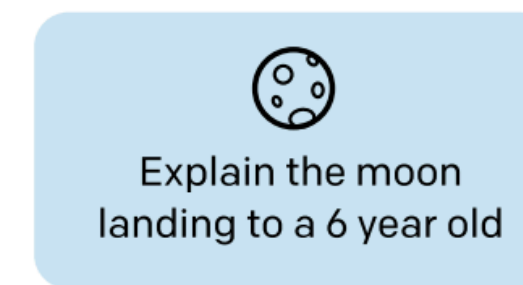
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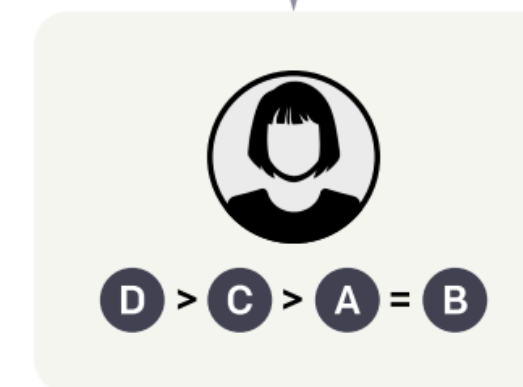
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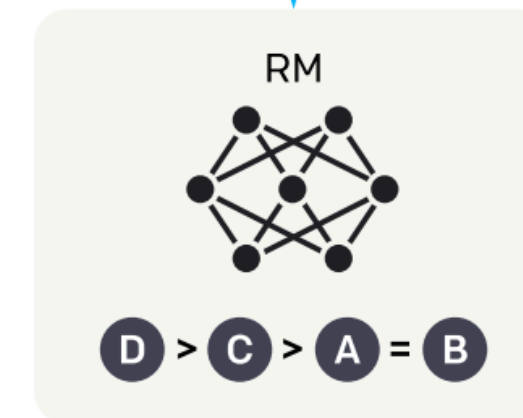
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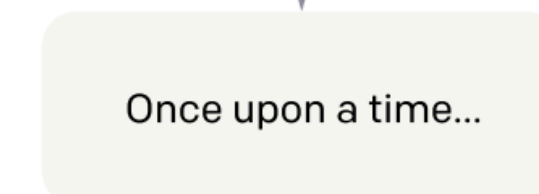
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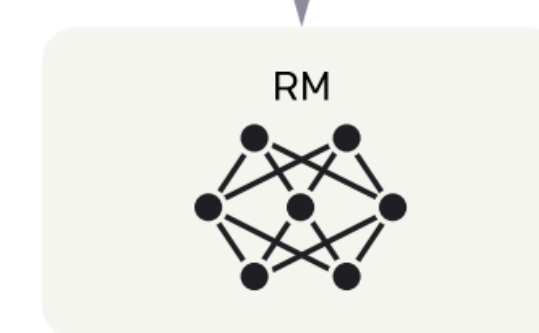
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Reward Model

The model learns to prioritize better answers.

$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} [\log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l)))]$$

K – number of ranked responses ($K \in [4, 9]$)

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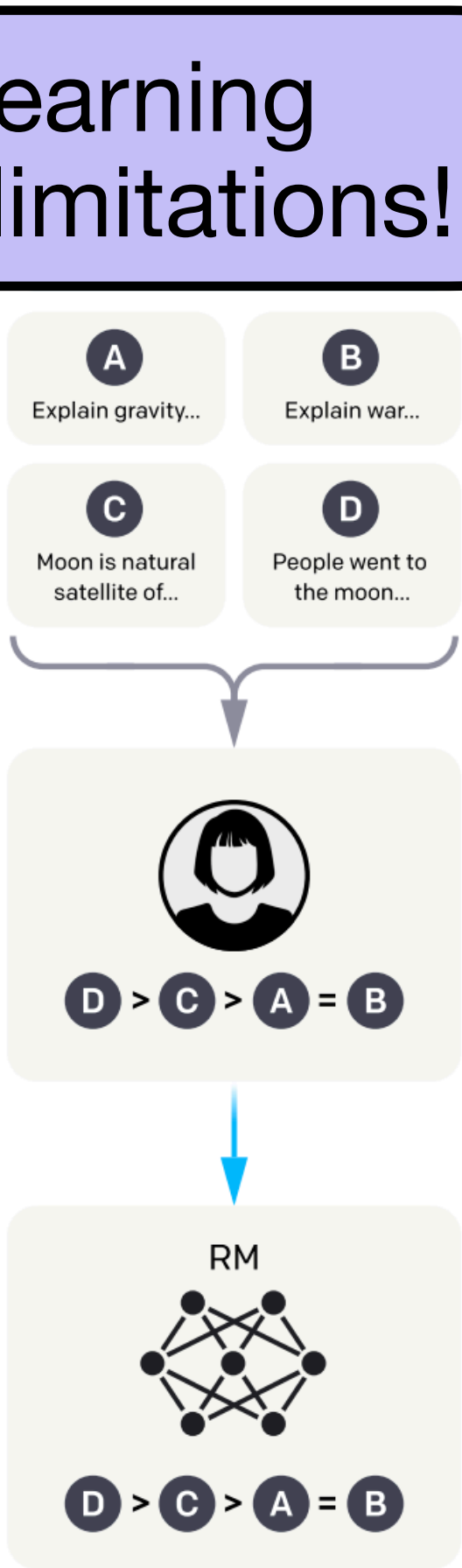
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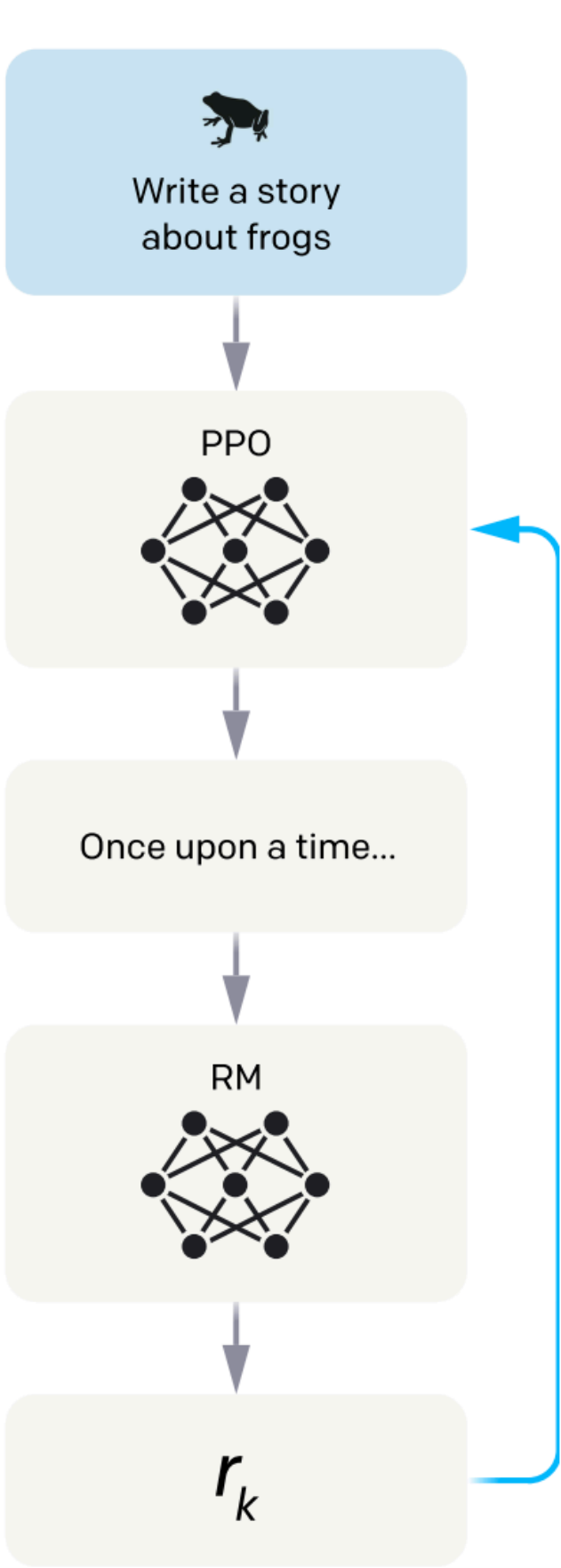
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Trust Region Policy Optimization (TRPO)

Current parameters

Previous parameters

Advantage $\hat{A}_t = r_t(s_t, a_t) - V_{\theta}(s_t)$

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t \right] \rightarrow \max$$

If $\hat{A}_t > 0$, then the chosen action is better than others on average

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We prohibit major changes to the policy subject to $\hat{\mathbb{E}}_t[\text{KL}[\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)]] \leq \delta$

$$\underset{\theta}{\text{maximize}} \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t - \beta \text{KL}[\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)] \right]$$

Proximal Policy Optimization (PPO)

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t \right] \rightarrow \max$$

Training stabilization

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(\underbrace{\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}}_{\text{Original objective}} \hat{A}_t, \underbrace{\text{clip} \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t}_{\text{Protects against huge updates (reward hacking)}} \right) \right]$$

Original objective

Protects against huge updates
(reward hacking)

Proximal Policy Optimization (PPO)

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t, \text{clip} \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right]$$

Final PPO objective

$$L_t^{CLIP+VF}(\theta) = \hat{\mathbb{E}}_t [L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta)]$$

$$\hat{A}_t = r_t(s_t, a_t) - V_{\theta}(s_t)$$

Value function loss: $(V_{\theta}(s_t) - V_t^{\text{targ}})^2$

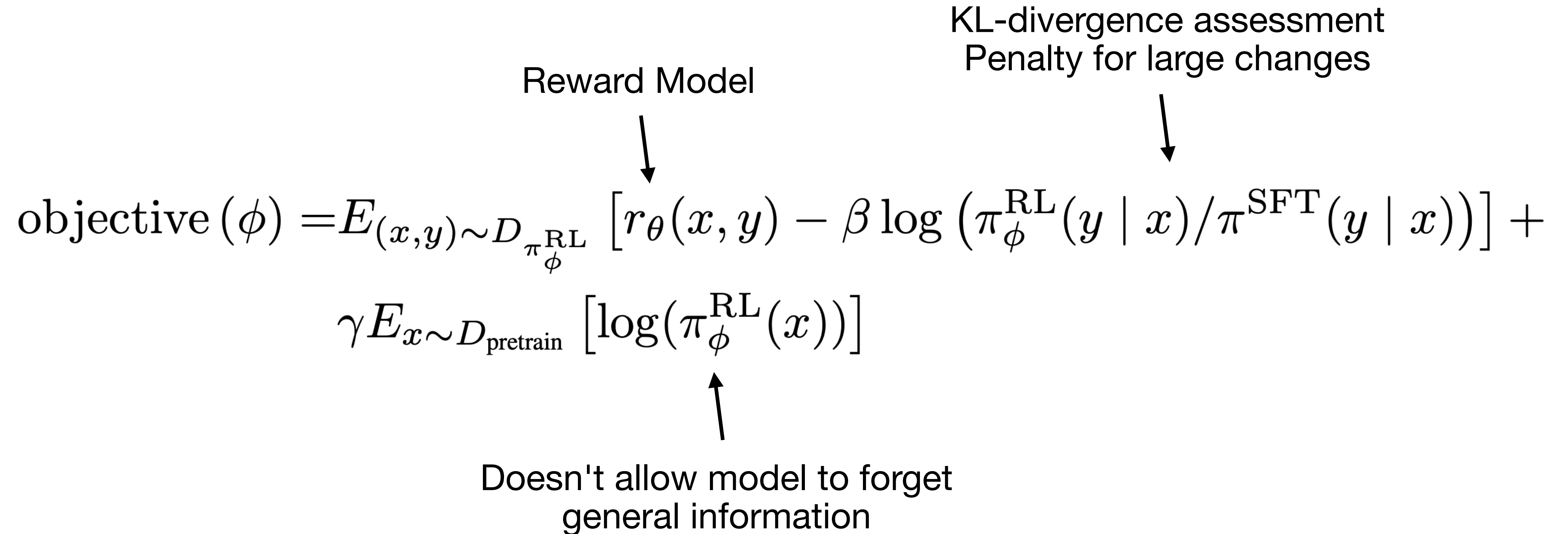
RL in RLHF

Reward Model

KL-divergence assessment
Penalty for large changes

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[r_{\theta}(x, y) - \beta \log \left(\pi_{\phi}^{\text{RL}}(y \mid x) / \pi^{\text{SFT}}(y \mid x) \right) \right] +$$
$$\gamma E_{x \sim D_{\text{pretrain}}} \left[\log(\pi_{\phi}^{\text{RL}}(x)) \right]$$

Doesn't allow model to forget
general information



RL in RLHF

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[r_{\theta}(x, y) - \beta \log \left(\pi_{\phi}^{\text{RL}}(y \mid x) / \pi^{\text{SFT}}(y \mid x) \right) \right] + \\ \gamma E_{x \sim D_{\text{pretrain}}} \left[\log(\pi_{\phi}^{\text{RL}}(x)) \right]$$

Value function training

$$L^{VF}(\xi) = \mathbb{E}_{(x,y) \sim D_{\pi_{\phi_t}^{\text{RL}}}} \left[\left(r_{\theta}(x, y) - V_{\xi}(x) \right)^2 \right]$$

Advantage $A(x, y) = r_{\theta}(x, y) - V_{\xi}(x)$

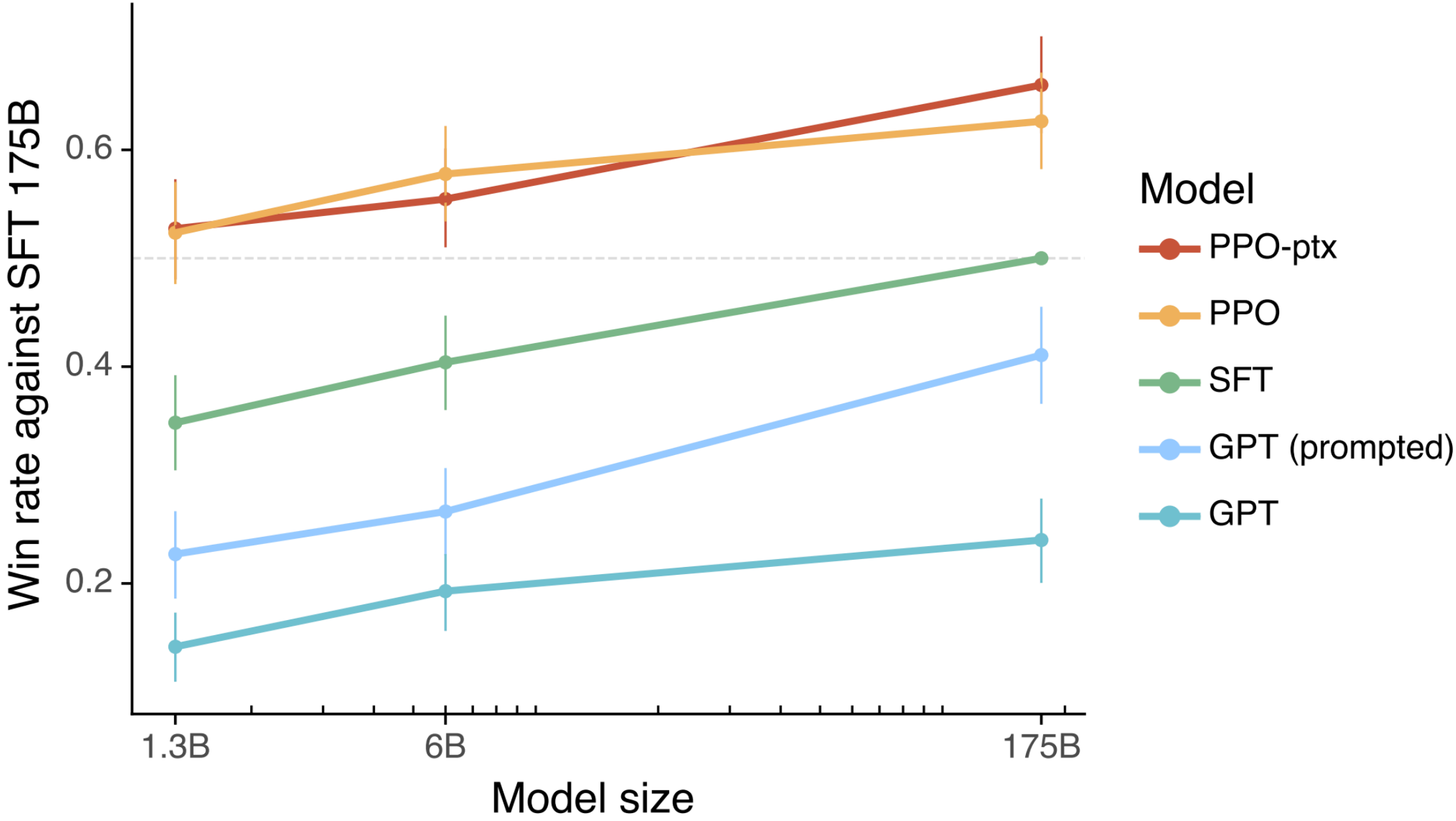
PPO for RL & RLHF

$$L_{\text{PPO-ptx}}(\phi) := E_{(x,y) \sim D_{\pi_{\phi_t}^{\text{RL}}}} \left[\min \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi_{\phi_t}^{\text{RL}}(y|x)} A(x,y), \text{clip} \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi_{\phi_t}^{\text{RL}}(y|x)}, 1 - \epsilon, 1 + \epsilon \right) A(x,y) \right) \right. \\ \left. - \beta \log \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)} \right) \right] + \gamma E_{x \sim D_{\text{pretrain}}} [\log(\pi_{\phi}^{\text{RL}}(x))]$$

RLHF results

Table 6: Dataset sizes, in terms of number of prompts.

SFT Data			RM Data			PPO Data		
split	source	size	split	source	size	split	source	size
train	labeler	11,295	train	labeler	6,623	train	customer	31,144
train	customer	1,430	train	customer	26,584	valid	customer	16,185
valid	labeler	1,550	valid	labeler	3,488			
valid	customer	103	valid	customer	14,399			



RLHF limitations

- The need to learn both **reward model** and **value function**

Is it possible not to use the **value function**?

Group Relative Policy Optimization (GRPO)

PPO:

$$L_{\text{PPO}}(\phi) := E_{(x,y) \sim D_{\pi_{\phi_t}^{\text{RL}}}} \left[\min \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi_{\phi_t}^{\text{RL}}(y|x)} A(x,y), \text{clip} \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi_{\phi_t}^{\text{RL}}(y|x)}, 1 - \epsilon, 1 + \epsilon \right) A(x,y) \right) - \beta \log \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)} \right) \right]$$

Monte Carlo estimation of KL divergence
(Large variance)

Group Relative Policy Optimization (GRPO)

PPO:

$$L_{\text{PPO}}(\phi) := E_{(x,y) \sim D_{\pi_{\phi_t}^{\text{RL}}}} \left[\min \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi_{\phi_t}^{\text{RL}}(y|x)} A(x,y), \text{clip} \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi_{\phi_t}^{\text{RL}}(y|x)}, 1 - \epsilon, 1 + \epsilon \right) A(x,y) \right) - \beta \log \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)} \right) \right]$$

Monte Carlo estimation of KL divergence
(Large variance)

GRPO:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} || \pi_{\text{ref}}) \right)$$

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}$$

We do not use the **value function**

$$\mathbb{D}_{\text{KL}}(\pi_{\theta} || \pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{\text{ref}}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1$$

Bregman divergence
(Lower variance)

More RLHF limitations

- The need to train a **reward model** and **value function**
- The quality of the model is limited by the quality of the **reward model**
- The **reward model** is limited by the sample size

Is it possible not to use the **reward model**?

Direct Policy Optimization (DPO)

The model is trained directly on **preference** data

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

DPO limitations

- DPO requires cleaner data
- DPO can lead to overfitting (collapse)
- Sensitivity to hyperparameters

Model	Method	Intro.	Inter.	Comp.
GPT-Neo 2.7B	SFT	5.6%	0.8%	0.0%
Codex 12B	SFT	9.7%	0.5%	0.1%
CodeT5	CodeRL	16.4%	4.9%	2.8%
AlphaCode 1B	5@1k	14.4%	5.6%	4.6%
Code Llama 7B	Few shot	10.8%	2.0%	0.8%
	SFT	30.0%	7.8%	2.8%
	DPO-Iter	20.9%	3.4%	1.3%
	PPO	29.4%	7.6%	2.4%
Code Llama 13B	Few shot	23.7%	5.6%	2.1%
	SFT	33.7%	8.7%	3.6%
	DPO-Iter	33.0%	8.0%	2.8%
	PPO	36.4%	11.47%	4.6%
Code Llama 34B	Few shot	32.8%	8.8%	2.9%
	SFT	38.6%	10.1%	3.9%
	DPO-Iter	34.2%	9.3%	3.7%
	PPO	44.4%	18.0%	9.1%

Reasoning

Chain of Thought

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. **X**

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. **✓**

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 **X**

(d) Zero-shot-CoT (Ours)

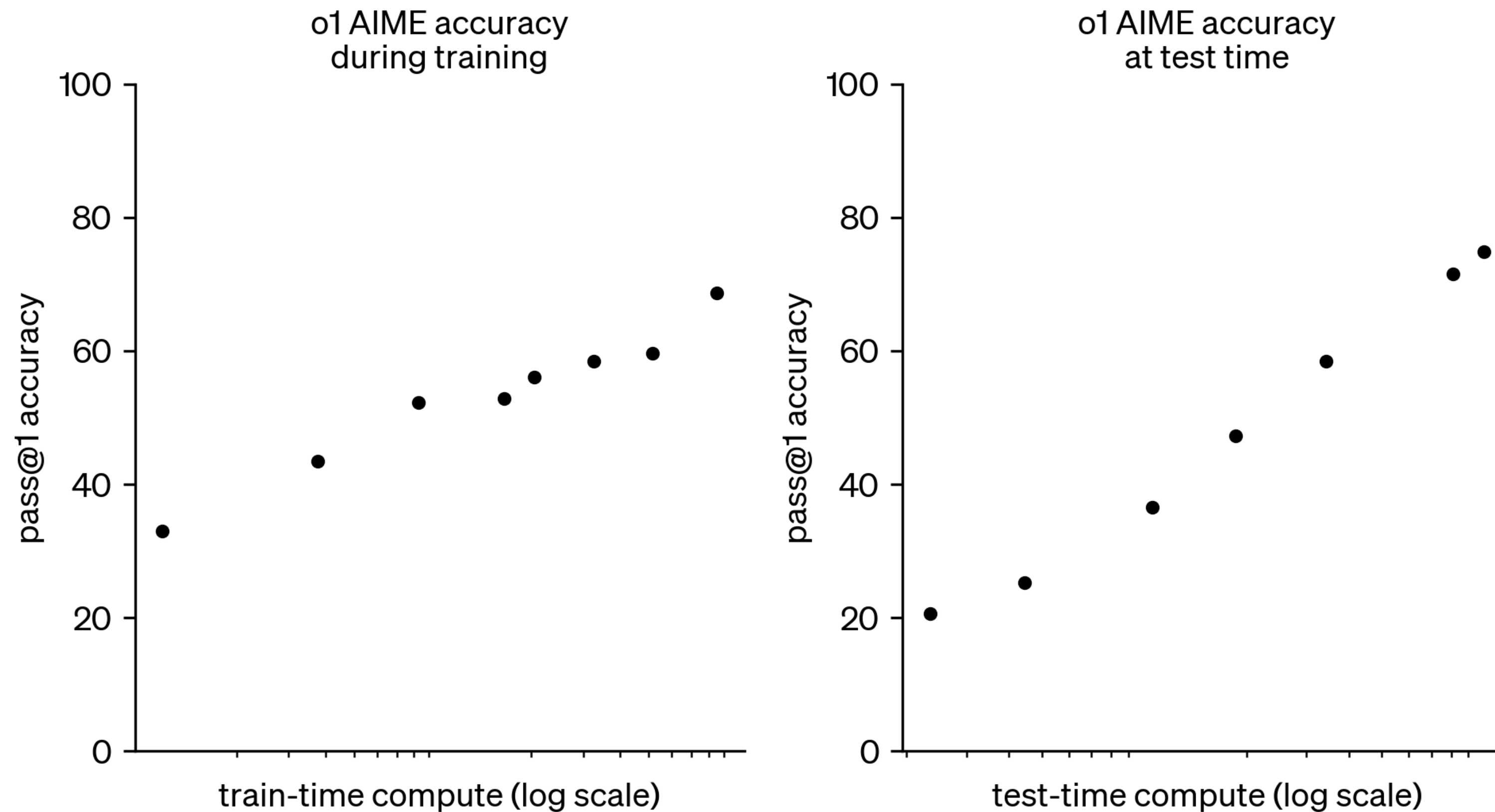
Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. **✓**

Test-time compute scaling

The more tokens a model generates, the higher its quality.



Thinking tokens

OpenAI o1 adds "thinking" tokens

Question: How can I poop without taking off my sweater?

<think> ... Considering all options ... </think>

Answer: The sweater doesn't interfere

Thinking tokens

OpenAI o1 adds "thinking" tokens

Question: How can I poop without taking off my sweater?

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Answer: The sweater doesn't interfere

How did they do it?

DeepSeek-R1-Zero

Open-source reasoning model trained without SFT and RLHF

1) Training with prompt

Pre-trained LLM (GPT, LLaMA) is trained to generate answers to a prompt

```
A conversation between User and Assistant. The user asks a
question, and the Assistant solves it. The assistant first
thinks about the reasoning process in the mind and then
provides the user with the answer. The reasoning process
and answer are enclosed within <think> </think> and
<answer> </answer> tags, respectively, i.e., <think>
reasoning process here </think>
<answer> answer here </answer>. User: {prompt}. Assistant:
```



A difficult question that requires thinking.

2) Rule-based reward

The reward is set according to two rules

1) Accuracy of the answer

4. Given the three sides of an obtuse triangle are 3, 4, x , then the range of values for x is ().

- (A) $1 < x < 7$.
- (B) $5 \leq x < 7$.
- (C) $1 < x < \sqrt{7}$.
- (D) $5 < x < 7$ or $1 < x < \sqrt{7}$.

Answer

D

2) Compliance with the format

```
<think>\nOkay, so I have this problem here where I need to find the range of values for x, given that the three sides of an obtuse triangle are 3, 4, and x. The answer choices are A through D. Let me think through this step by ...</think>
```

We check that all the "thinking" is where it should be

Training

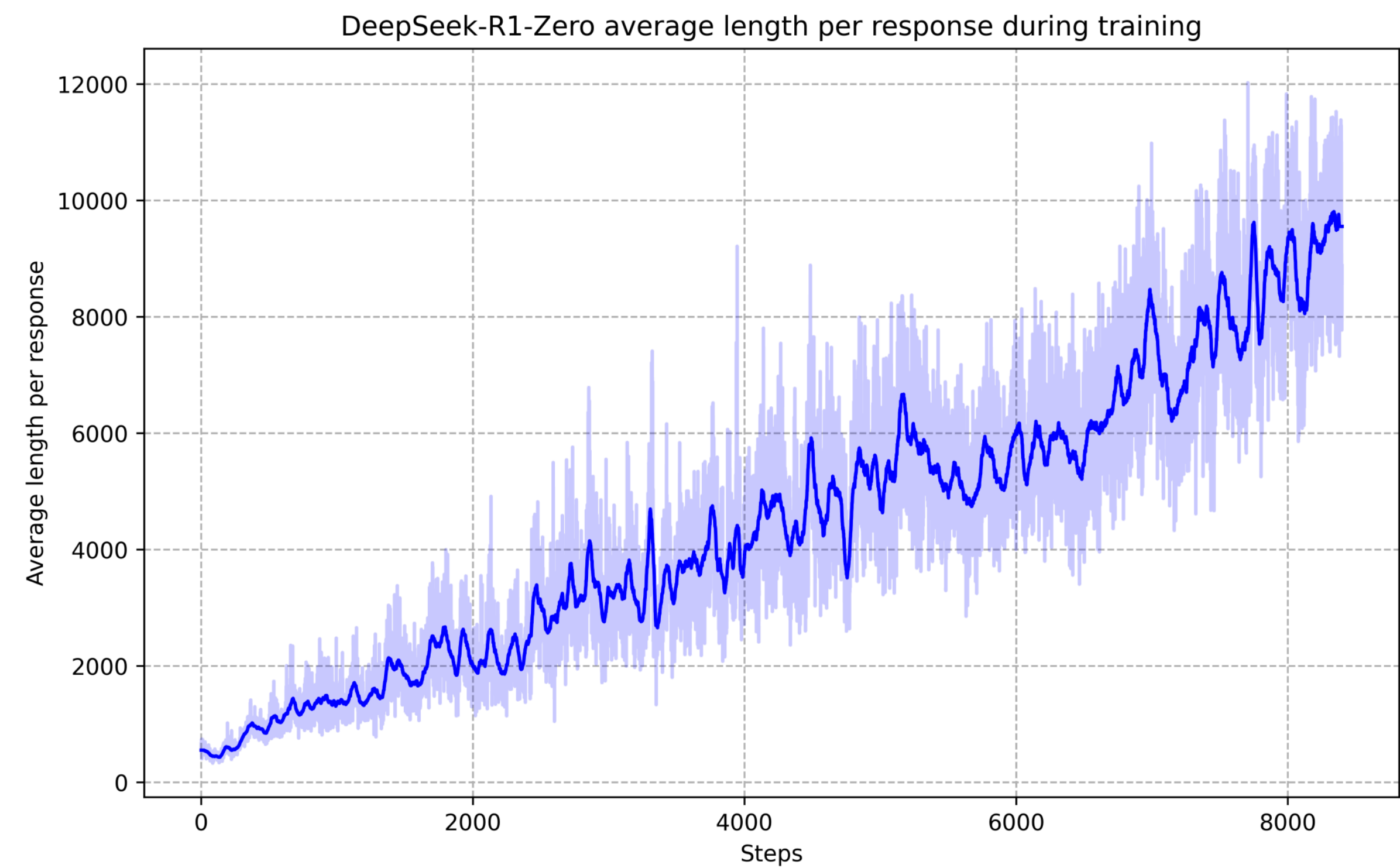
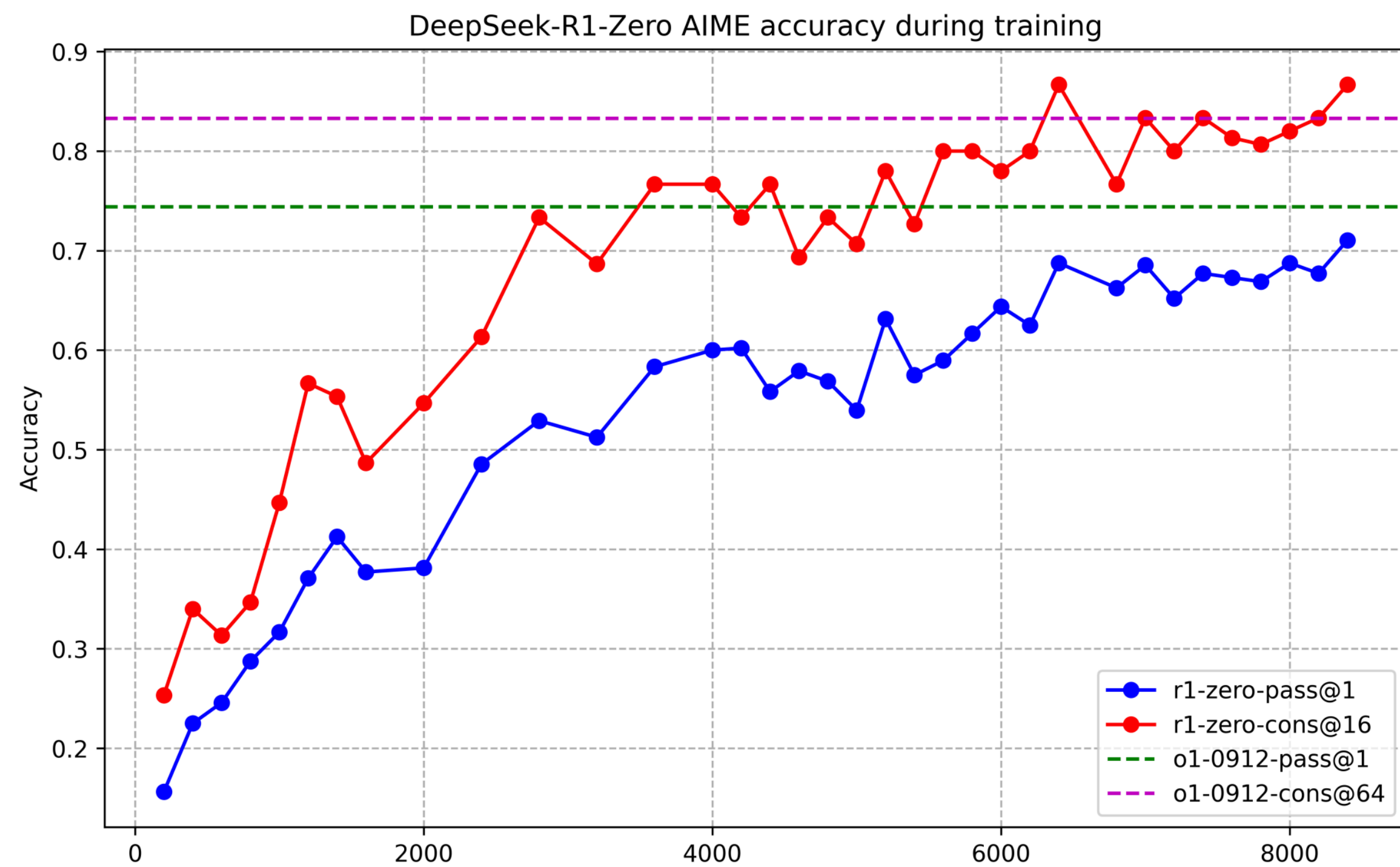
The model is trained using GRPO because it does not require any additional models.

$$\mathcal{J}_{GRPO}(\theta) = \frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \varepsilon, 1 + \varepsilon \right) A_i \right) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right)$$

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}$$

$$\mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1$$

Results



Results

The model herself learned to think correctly

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

Limitations

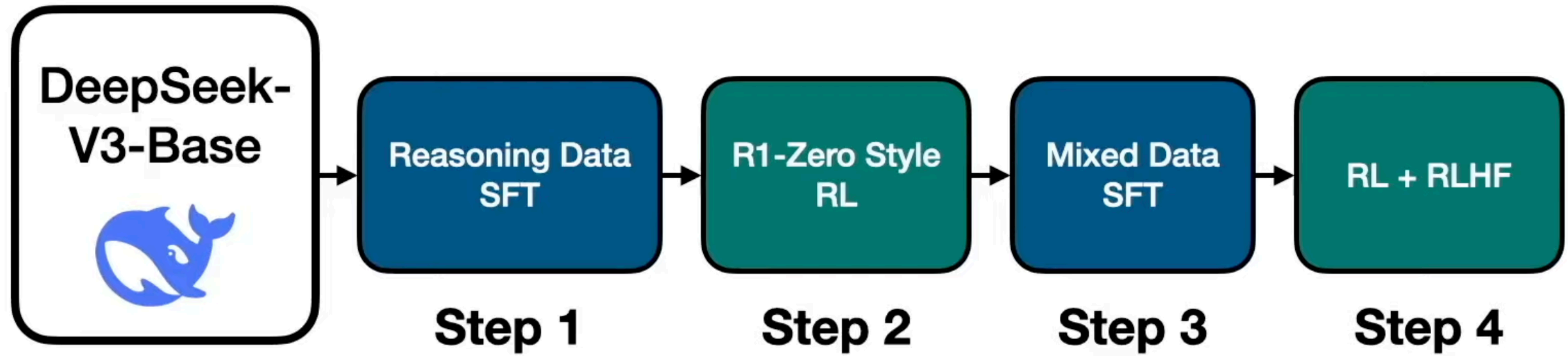
Readability

The model doesn't always respond in the required format.

Language Mixing

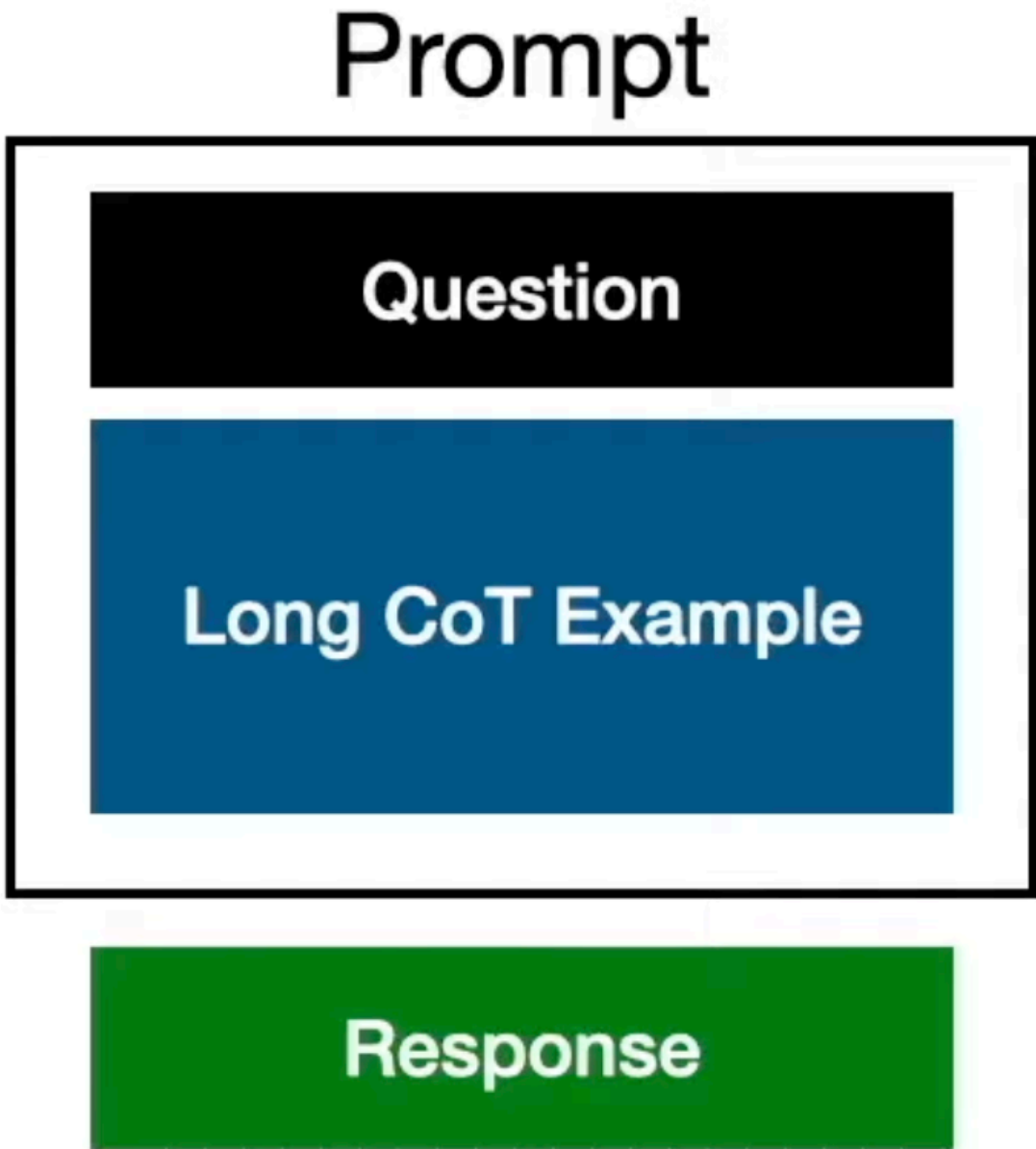
The model responds in the wrong language.

DeepSeek-R1

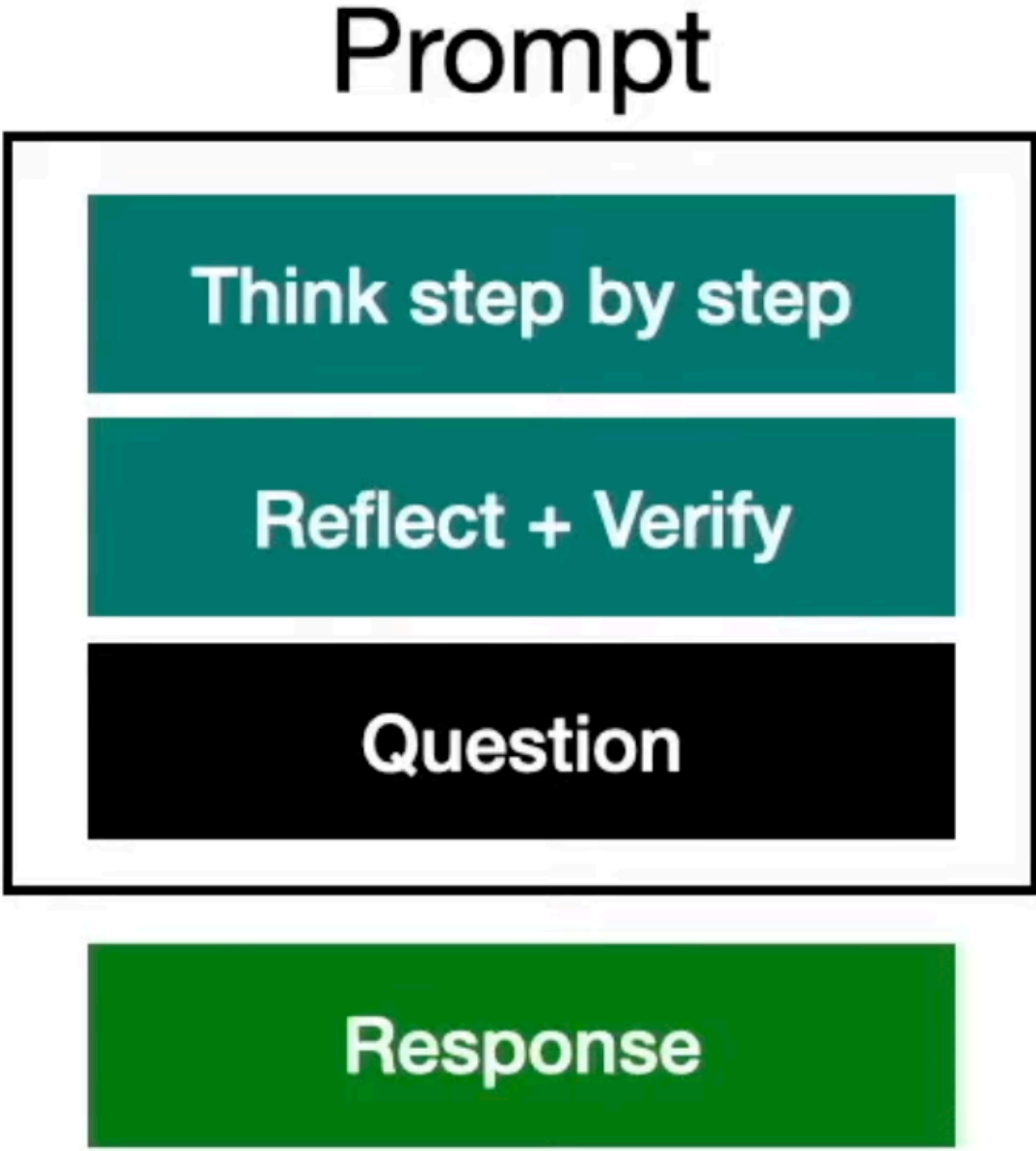


Step-1: SFT + reasoning

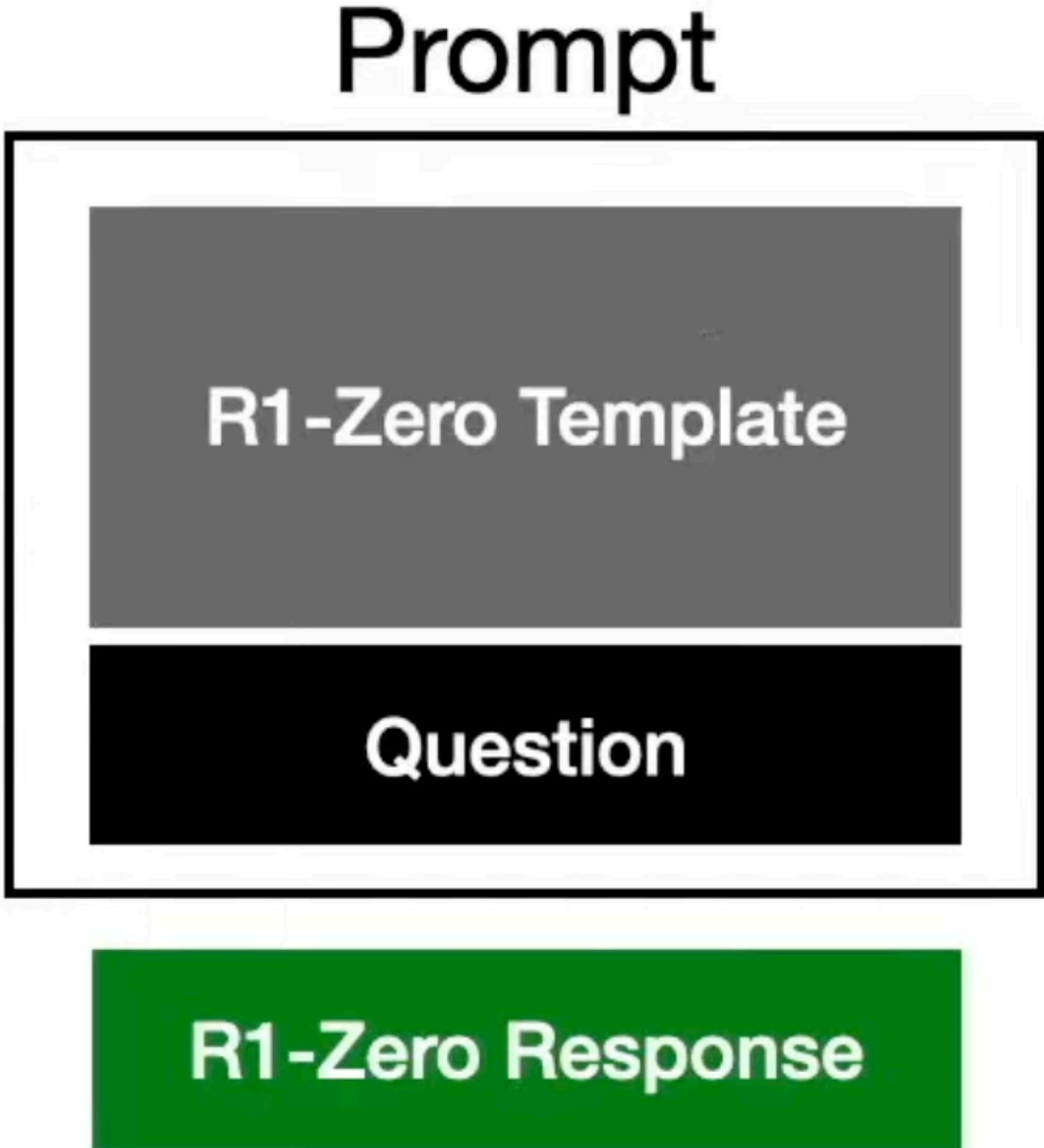
Training the model to respond in the required format



Examples with reasoning



Examples with chain-of-thought



Examples from R1-Zero

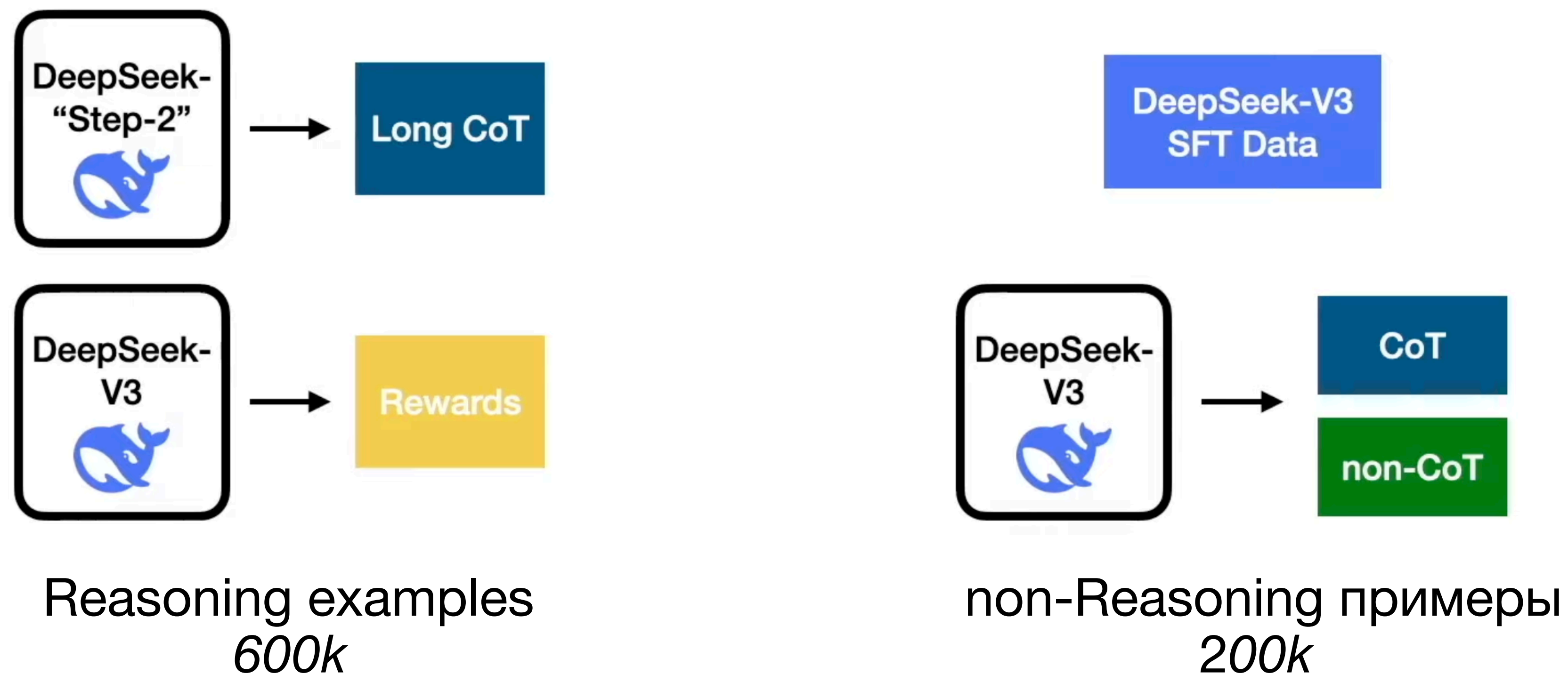
Step 2: RL like in R1-Zero

Reward for:

- 1) Answer accuracy
- 2) Format compliance
- 3) Language compliance

Step-3: SFT on mixture of data

At the moment, the model thinks every time

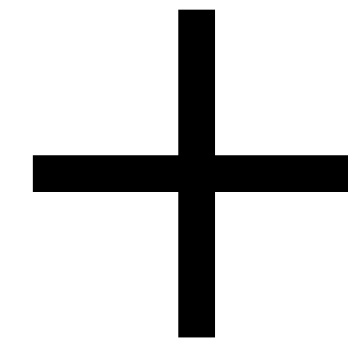


Step-4: Alignment (RL + RLHF)

Reward for:

- 1) Answer accuracy
- 2) Format compliance
- 3) Language compliance

R1-Zero-like



Reward for:

- 1) Usefulness
- 2) Safety

RLHF

Instead of a conclusion

Currently, the quality of LLM is limited by the size of the training set.

RL allows these limitations to be removed, as the agent can learn indefinitely in the environment.

